int a ← attribute

local variables —allocated runtime→ onto the stack.

3 logical spaces

→ BDS (global data seg.) all global variables COMP. T.

→ stack local variables RUNTIME

→ heap dynamic variables. RUNTIME

p ← pointer = malloc(how_many_bytes / sizeof(type))

A dynamic variable doesn't have a name. EVER.

Structural:

variable = (name, set_of_ attributes, reference (address), value)

→ data type

→ scope (DoV) - Visibility Domain (spatial) - portion of source code where the var. could be used.

→ Extend - Lifetime (temporal)

→ Memory class (auto, register, static, extern) stop in plain text exporting

C/C++

`static int a ;` => makes it local (for a global variable)

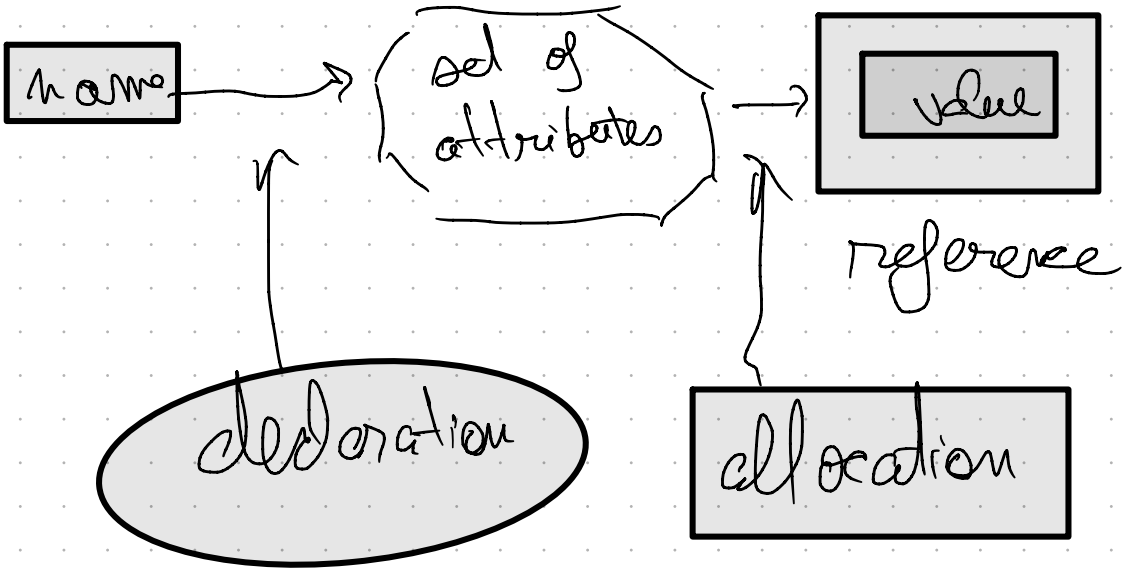
FOR A GLOBAL ONE. => you cannot export it.

`float eval(...) {` => local to just a function

`static int n = 17;` => lifetime is the whole program.

`}`

↳ local variable that keeps its value between calls.



Look at data definition directives PPT.

a1 db 0,1,2,'xyz'

a2 times db ?? h;

ah1 db a2+1; OBJ. can only handle 16/32 bits forst.

ah1 dw a2+1 ; offset of a2+1

≃ g